CHAPTER 2

THEORY OF CONSUMER BEHAVIOUR & EQUILIBRIUM

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 67

Learning Objectives

The students will learn:

- *Law of Diminishing Marginal Utility and the relationship between Total and Marginal Utility.*
- *Indifference Curve, its meaning and characteristics.*
- *How consumer attains equilibrium by marginal utility, Equi-marginal utility and Indifference Curves.*

Topic-1
Utility and Cardinal Theory **Page No. 22**
Topic-2
Ordinal Theory of Utility **Page No. 28**

Topic-1 Utility and Cardinal Theory

Concepts covered: Meaning of utility, Cardinal approach of utility, Types of utility- Marginal and Total Utility, Law of diminishing Marginal Utility- Meaning, Schedule and Diagram, Consumer's equilibrium - One commodity case, Law of Equi-Marginal Utility.

Revision Notes

- **Consumer:** The individual who takes the decision to buy a certain commodity for satisfaction of his needs or the needs of the family.
- **Utility:** Utility is the power or capacity of a commodity to satisfy human wants. Alternatively, utility of a commodity means the amount of satisfaction that a person gets from consumption of a good or service.
- **Cardinal Theory of Utility:** It is the theory according to which utility is measured in terms of numbers like 2, 4, 6, 8, etc. Cardinal utility is measured in terms of units (the units on a scale of utility or satisfaction). According to cardinal utility the goods and services that are able to derive a higher level of satisfaction to a consumer will be assigned higher utility and goods that result in a lower level of satisfaction will be assigned lower utility. Cardinal utility is a quantitative method that is used to measure consumption satisfaction.
- **Total Utility:** Total utility is the total psychological satisfaction a consumer obtains from consuming a given amount of a particular goods. It is the sum total of utility derived from the consumption of all the units of a commodity.

$$TU = MU_1 + MU_2 + MU_3 + \dots\dots\dots\dots + MU_n \text{ or } TU = \Sigma MU_n$$

- **Marginal Utility:** It refers to additional utility on account of the consumption of an additional unit of a commodity. It is calculated as:

$$MU = TU_n - TU_{(n-1)} \text{ or } MU = \frac{\Delta TU}{\Delta Q}$$

- **Relationship between Total Utility and Marginal Utility:**
  - When MU is positive, TU is increasing.
  - When MU is zero, TU is maximum.
  - When MU is negative, TU is decreasing.

Scan to know more about this topic
Relationship between TU and MU

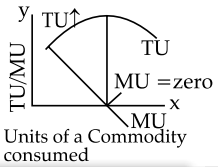
- **Law of Diminishing Marginal Utility:** Law of diminishing marginal utility states that as more and more units of a commodity are consumed, marginal utility derived from every additional unit must decline. This law is based on these assumptions:
  - The utility can be measured in terms of units.
  - The marginal utility of money is constant.
  - The consumption must be in identical and standard units.
  - The consumption is continuous.
  - The consumer is rational.

Theory of Consumer Behaviour & Equilibrium

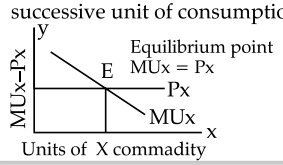
Cardinal Approach

Utility Analyse

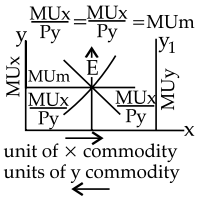
- Utility want satisfying power of a commodity
- Total Utility: $TU = \Sigma MU$
- Marginal Utility:
  - $MU = TU_n - TU_{n-1}$
  - $MU = \frac{\Delta TU}{\Delta Q}$
- Relationship between TU and MU
  - When MU is positive, TU increases
  - When MU is zero, TU is maximum
  - When MU is negative, TU declines



- Law of Diminishing Marginal Utility (Gossen's first law)
  - Statement: Law of Diminishing Marginal Utility states that with every additional unit of consumption of a commodity, its MU falls.
  - Assumptions
    - Homogeneous units of commodity consumed
    - No time gap in consumption of commodity
    - MU of money remains constant
    - No change in price of commodity
    - No change in tastes, habits and income of consumer
    - Consumer is Rational
    - No change in price of related goods
  - Exceptions of Law of Diminishing Marginal Utility
    - Very small size of units of consumption
    - Alcoholic drinks
    - Miser man
    - Rare commodities
    - Music
    - Hobbies of collecting stamps, old coin, etc.
- Determination of consumer equilibrium
  - Assumptions
    - Utility can be measured in cardinal numbers
    - Consumer is rational
    - Consumer's income is given and remains constant
    - Prices of commodities are given
    - Law of Diminishing marginal utility operates
  - Conditions of Consumer Equilibrium
    - In case of single commodity
      1. $MUx = Px$
      2. MUx declines with every successive unit of consumption



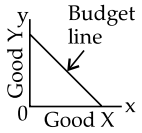
    - In case of two or more commodities
      - Equilibrium determined by the Law of Equi Marginal Utility
        1. $\frac{MUx}{Px} = \frac{MUy}{Py} = MUm$
        2. Expenditure = Income
           $Px . Qx + Py . Qy$ = Money income.
        3. MU falls as more of a good is consumed.



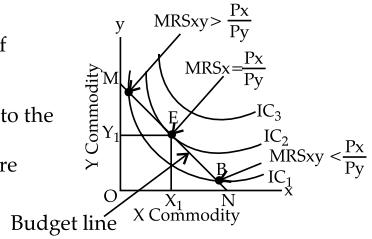
Ordinal Approach

Consumer's Equilibrium

- Assumptions
  1. Consumer is Rational
  2. Ordinal measurement
  3. Transitivity of choice
  4. Diminishing marginal rate of substitution (Rate of sacrifice of amount of Y commodity diminishes with every additional unit of X commodity).
- Indifference Curve
  - Meaning: IC shows various combinations of two commodities which give equal satisfaction to the consumer.
  - Properties of Indifference Curve
    1. IC slopes downwards (Left to right)
    2. IC is convex to its origin due to DMRSxy.
    3. Two ICs never intersect each other
    4. Higher IC represents higher level of satisfaction.
  - Indifference map: Set of different ICs, each of which represents a different level of satisfaction
- Budget line/Price line
  - Meaning: Shows various combinations of two goods, A consumer can buy at given prices of two goods with his given income.
  - Features
    - Downward sloping straight line curve
    - Slope of budget line = Px/Py
    - Budget line changes, if Px, Py or Income changes



- Consumer's Equilibrium
  - $MRSxy = Px/Py$ (slope of IC = slope of budget line)
  - Diminishing MRSxy (IC should be convex to the origin)
  - Income = Expenditure
    $M = Px . Qx + Py . Qy$



Trace the Mind Map
First Level · Second Level · Third Level

- The income of the consumer and the prices of goods are constant.

➢ **Law of Equi-Marginal utility:** The law of equi-marginal utility states that the consumer will distribute his money income between the goods in such a way that the utility derived from the last rupee spent on each good is equal.

➢ **Consumer's Equilibrium:** The consumer is in equilibrium when, given his income and market prices, he plans his expenditure on different goods and services, in such a manner that he maximises his total satisfaction.

➢ **Consumer Equilibrium in case of a Single Commodity:** A consumer purchasing a single commodity will be at equilibrium, when he is buying such a quantity of that commodity, which gives him maximum satisfaction. Being a rational consumer, he will be at equilibrium when marginal utility is equal to the price paid for the commodity i.e.,

$$\frac{MU_x}{P_x} = MU_m$$

For example, if the price of a shirt is ₹ 400, the consumer will be in equilibrium when he attains satisfaction equal to ₹ 400.

Scan to know more about this topic

Consumer's equilibrium

➢ **Consumer Equilibrium in case of Two Commodities:** A consumer purchasing two commodities will be at equilibrium when he spends his limited income in such a way that the ratios of marginal utilities of two commodities and their respective prices are equal and MU falls as consumption increase i.e.,

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} = MU_m$$

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State the meaning of consumer equilibrium? [OEB]

Ans. Consumer's equilibrium can be defined as when consumer spends on the consumption of a commodity such that he derives maximum satisfaction.

Q.2. Give the difference between Cardinal utility approach and Ordinal utility approach? [OEB]

Ans. Cardinal utility approach is based on using utility as the parameter whereas Ordinal utility approach is based on rank parameter.

Q.3. Define marginal utility? [OEB]

Ans. Marginal utility can be defined as the increase in total utility with the increase in consumption of units of a commodity.

Q.4. State whether the given statements are true or false. Give the reason for the answer. [OEB]

(a) "As the consumption of a commodity increases, then there is rise in marginal utility of the commodity."

Ans. False

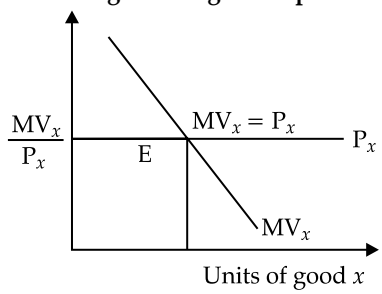
Reason: The given statement is false as with the increase in the units of consumption, marginal utility falls of the commodity. Because with the rise in consumption, total utility rises upto a limit but MU falls.

(b) "When total utility starts falling, then marginal utility becomes zero".

Ans. False

Reason: When total utility starts falling, marginal utility becomes negative. And when total utility is maximum, then marginal utility is zero.

Q.5. What does the given diagram depict? [OEB]



Ans. The above diagram depicts consumer equilibrium using one commodity case. As price is constant and parallel to x-axis. Marginal utility curve is downward sloping. And the point where price equals to marginal utility, there equilibrium is attained.

2 Marks Questions

Q.1. State the law of Diminishing Marginal Utility. Mention any two assumptions of the law.

[ISC 2020]

Ans. Law of diminishing marginal utility:

The law states that as the amount consumed of a commodity increases, the other things being constant, **the utility derived by the consumer from the additional units,** i.e., marginal utility goes on decreasing.

Two assumptions of the law:

- All units of the commodity must be identical.
- Unit of the good must be standard.
- There should be no change in taste during the process of consumption.
- There must be continuity in consumption.
- There should be no change in the price of substitute goods.

- Utility is measurable.
- Consumer is rational.
- MU of money is assumed to be constant.

(Any Two) [ISC Marking Scheme, 2020]

Examiner's Comment

- Majority of the candidates while explaining the Law of DMU, instead of stating that it is marginal utility/ additional utility / additional benefit which decreases, wrote that utility decreases with increase in consumption / with consumption of successive units of commodity / with consumption of additional units of a commodity. They did not realize that this implies a decline in total utility. Most candidates did not write the words additional units while explaining the meaning of the Law. Many candidates wrote that money income remains constant as an assumption of the law while this assumption is not related with the Law of DMU.

Answering Tip

- Students should write 'other things being constant' and also 'additional units' while explaining the meaning of Law of DMU. Along with this, the assumptions should also be learnt in detail. It should be emphasized that the Law of DMU cannot be written in the form of an example or table which is used to only explain the concept.

Q.2. Define total utility. How is marginal utility derived from total utility? [ISC 2018]

Ans. Total utility is the total amount of satisfaction a consumer gets by consuming.

OR

It refers to the total satisfaction derived by the consumer from the consumption of a specific quantity of a commodity.

$MU_n = TU_n - TU_{n-1}$ as marginal utility is the satisfaction derived from the additional unit consumed.

Examiner's Comment

- Majority of the candidates defined total utility as utility derived from all commodities. Many candidates wrote TU=∑MU or simply defined utility.

Answering Tip

- Understand that TU=∑MU is a relation between TU and MU whereas total utility is defined as the utility obtained from consumption of certain or specific quantity of a commodity.

Q.3. Define Marginal utility. When can it be negative? [ISC 2015]

Q.4. Complete the following table: [OEB]

Units consumed	TU	MU
0	0	—
1	12	—
2	—	10
3	29	—

Ans.

Units consumed	TU	MU
0	0	—
1	12	12
2	22	10
3	29	7

Q.5. If MUx = 8, and Px = 5, then what happens to the consumer? Is the consumer in equilibrium? [OEB]

Ans. Since MUx(8) > Px(5), hence marginal utility derived from the consumption of a goods is greater than the price of a goods. Hence, consumer consumes more of a goods and thus MU falls till the point where both MU and price are equal.

Q.6. Complete the following table:

Units consumed	TU	MU
0	0	—
1	15	—
2	—	12
3	35	—
4	40	—

Ans.

Units consumed	TU	MU
0	0	—
1	15	15
2	27	12
3	35	8
4	40	5

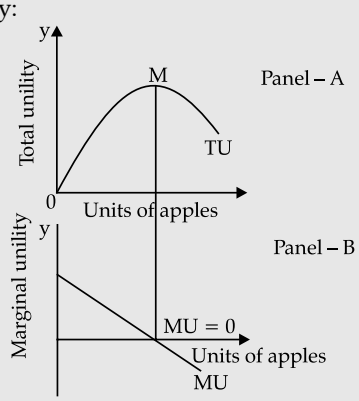
This question is for practice and its solution is available at the end of the chapter

3 Marks Questions

Q.1. Explain with the help of a diagram the relationship between total utility and marginal utility.

[ISC 2017]

Ans. The relationship between total utility and marginal utility:



(i) The total utility curve is **concave** from above indicating declining slope of the curve, which indicates a declining marginal utility. Thus up to **point M, TU curve has a positive slope**, but its slope goes on decreasing steadily as quantity consumed is increased. This shows that **marginal utility decreases** but is positive as long as total utility increases.

(ii) When **total utility is maximum marginal utility is zero**.

(iii) After M, the slope of **TU curve becomes negative showing that MU is negative.** Thus when total utility declines marginal utility is negative.

[ISC Marking Scheme, 2017]

Examiner's Comment

- Most of the candidates did not show the relation between the maximum point of TU curve and the point of intersection of MU curve. Some candidates wrongly drew TU curve as TP curve.

Answering Tip

- Learn the relationship with the help of diagrams and practice diagrams.

Q.2 The price of a book is 500. Using the given table identify the where the consumer will attain equilibrium. **[OEB]**

Number of books	MU
1	700
2	600
3	500
4	400
5	300

Ans. The essential condition satisfied for a consumer to be in equilibrium in a single commodity market is $\frac{MU_x}{P_x} = MU_m$. If MU_m is assumed to be 1.

As shown in the table it can be observed from the slope of the graph, the MU derived from purchase of each additional unit of books is decreasing.

As per the condition stated above, the consumer is in equilibrium where $MU_x = P_x$. In the diagram, this condition is attained where price of the books is same as the MU derived from the purchase of books, that is, 500. So, the point of consumer's equilibrium is 3 units of books.

Q.3. When $MU_x/P_x > MU_y/P_y$, then what actions are taken by the consumer to reach equilibrium? **[OEB]**

Ans. Consumer attains equilibrium when $MU_x/P_x = MU_y/P_y$.

But when $MU_x/P_x > MU_y/P_y$, it means that per rupee marginal utility derived from consumption of an additional unit of a commodity X is greater than per rupee marginal utility derived from consumption of commodity Y. Hence, consumer consumes more of X and less of goods Y. Hence, MU of X falls and that of Y rises till the point where both marginal utilities are equal.

Q.4. If $MU_x = 4, P_x = 2$ AND $MU_y = 8, P_y = 4$, then is the consumer in equilibrium? **[OEB]**

Ans. Here, condition to attain consumer equilibrium is:

$$MU_X/P_X = MU_Y/P_Y$$
$$4/2 = 8/4$$
$$2 = 2$$

Since per rupee marginal utility derived from consumption of both the goods X and Y are equal. Hence, consumer is in equilibrium.

6 Marks Questions

Q.1. With the help of a schedule, explain the operation of the Law of Equi-Marginal Utility.

[ISC Specimen Paper, 2023-24]

OR

A consumer consumes goods X and Y. Given below is his marginal utility schedule for goods X and Y. Suppose, the price of X is ₹ 2, Y is ₹ 1 and income ₹ 12. State the law of Equi-marginal utility and explain how the consumer will attain equilibrium.

[ISC 2018]

Units	1	2	3	4	5	6
MU_X	16	14	12	10	8	6
MU_Y	11	10	9	8	7	6

Ans. Given below is an individual's Marginal Utility schedule for goods X and Y. Suppose, price of X is ₹ 2, Y is ₹ 1 and income ₹ 12.

Units	MU_X	MU_Y	MU_x/P_x	MU_y/P_y
1	16	11	8	11
2	14	10	7	10
3	12	9	6	9
4	10	8	5	8
5	8	7	4	7
6	6	6	3	6

The consumer will be in equilibrium when $MU_x / P_x = MU_y/P_y = MU$ per unit of money.

Combinations:

1 unit of X + 4 units of Y = ₹ 2 + ₹ 4 = ₹ 6

2 units of X + 5 units of Y = ₹ 4 + ₹ 5 = ₹ 9

3 units of X + 6 units of Y = ₹ 6 + ₹ 6 = ₹ 12

Since the budget is ₹ 12, the consumer will be in equilibrium when he buys 3 units of good X and 6 units of good Y. **[ISC Marking Scheme, 2018]**

Examiner's Comment

- Many candidates did not write the statement of the Law of Equi-marginal utility. A number of candidates did not state the equilibrium conditions for the two commodities. Many candidates were unable to write the correct combinations to show the consumer's equilibrium.

Answering Tips

- Learn the law of Equi-marginal utility showing $MU_x/P_x = MU_y/P_y = MU$ per unit of money.
- The students need to understand the correct combinations where equilibrium conditions are fulfilled and the reasons why some conditions are not correct.

Q.2. A consumer can attain a stable equilibrium only when the marginal utility obtained from a commodity can be equated with the price of that commodity. Explain this phenomenon with the help of a diagram.

[ISC Specimen Paper, 2023-24]

OR

With the help of a diagram, explain how the consumer will attain equilibrium on consumption of a single commodity at a given price. [ISC 2023]

OR

Explain with the help of a diagram the consumer's equilibrium through utility approach. [ISC 2015]

Ans. Equilibrium through utility approach:

Consumer's equilibrium means that consumer is in equilibrium when, given his income and market prices, he plans his expenditure on different goods and services in such a manner that he maximises his total satisfaction.

Condition for Equilibrium:

A consumer purchasing a single commodity will be at equilibrium when he is buying such a quantity of that commodity which gives him maximum satisfaction. Being a rational consumer, he will be at equilibrium when marginal utility is equal to the price paid for the commodity, i.e.,

$$\frac{MU_x}{P_x} = MU_m$$

MU_x - Marginal utility of commodity X

P_x - Price of commodity X

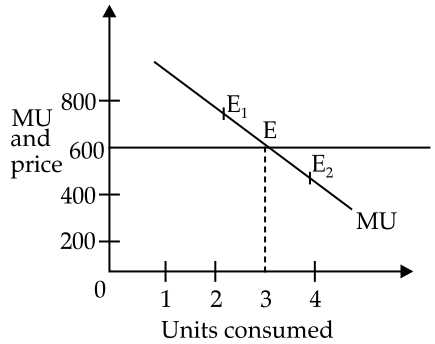
MU_m - Marginal utility of Money

Numerical example:

Price of shirt ₹ 600

Schedule:

Qty	MU_X
1	700
2	650
3	600 - Equilibrium
4	500



Consumer is in equilibrium when he buys 3 shirts as $MU_x = P_x$

When he buys 2 shirts $MU_x > P_x$, so he buys more.

When he buys 4 shirts, $MU_x < P_x$.

At E_1, $MU_x > P_x$, consumer buys more.

E_2, $MU_x < P_x$, buys less.

Examiner's Comments

- A number of errors were observed in this question:
- Some candidates used law of equi-marginal utility to explain Consumer's equilibrium;
- A few candidates explained this equilibrium using ordinal utility/ indifference curve approach.
- The explanations given by many candidates were devoid of unstable situations like effects of $MU_x > P_x$ and $MU_x < P_x$. Some candidates did not show it with $MU_x = P_x$.

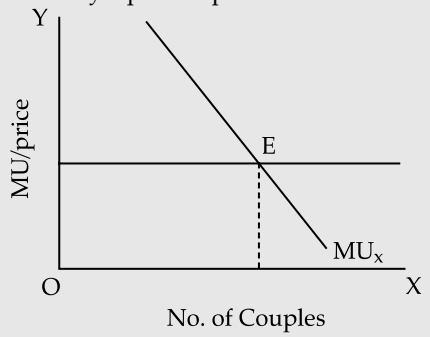
Answering Tips

- Learn the application of the different utility approaches. Understand clearly the differences between cardinal and ordinal approach of utility.
- Students should be learn that equilibrium means stable equilibrium. Therefore 'how the consumer attains the stable situation for unstable', is a must in the explanation.

Q.3. The marginal utility schedule of a rational consumer is given below. If the price of a commodity is ₹ 35, explain with the help of a diagram, how the consumer attains equilibrium. [ISC 2020]

Number of the commodity bought	1	2	3	4	5
MU	50	45	40	35	30

Ans. A utility maximizing consumer will be in equilibrium when he purchases that much quantity of the commodity where the MU of the commodity equals its price.



At E, equilibrium condition is satisfied when the consumer buys 4 units of the commodity i.e.,

$$MU_X = P_X.$$

Diagram should be as per the given schedule.

- If $MU_X > P_X$, i.e., before E, consumer gains more utility than price he pays. So, he continues consumption.
- If $MU_X < P_X$, i.e., beyond E, he has to pay more than his gain in utility, therefore he reduces his consumption.

So, equilibrium is attained at E where $MU_X = P_X$.

[ISC Marking Scheme, 2020]

Examiner's Comment

- Majority of the candidates did not write the equilibrium condition. Most candidates did not draw the diagrams as per the schedule. A large number of candidates only wrote that MU = P and at 4 units. They did not write what happens when MU > P and MU < P. Many candidates did not extend the price line beyond the equilibrium point.

Answering Tip

- Students should write the statement of the consumer's equilibrium, condition of the equilibrium, and then draw the graph and the schedule. They need to know that if the schedule is given then the graph should be based on it and three points to be explained like: What happens when MU > P, MU < P and MU = P.

Topic-2

Ordinal Theory of Utility

Concepts covered: Ordinal Approach of Utility, Concept of Indifference curves, indifference set and indifference map and monotonic preferences, Concept of Marginal rate of substitution. and Law of diminishing Marginal rate of substitution, Concept of Budget set, budget line, Consumer's equilibrium through Budget line and indifference curve approach.

Revision Notes

- **Ordinal Utility Approach:** This approach was laid down by Professor Allen and Hicks. According to this theory, utility cannot be measured in quantitative terms. Goods can only be compared with each other in terms of providing more utility, less utility or same utility. But it is rankable.
- **Monotonic Preferences:** Consumer's preferences becomes monotonic if the consumer between various bundles of two goods, prefers the bundle which has more of at least one of the goods and no less of other goods as compared to the other bundle. Monotonic preference means that a rational consumer always prefers more of a commodity as it offers him a higher level of satisfaction.
- **Indifference Set:** It is a set of those combinations of two goods which offer the consumer the same level of satisfaction, so that the consumer is indifferent across any number of combinations in his indifference set.

- **Indifference Curve:** A curve which is a graphical presentation of an indifference set showing different combinations of two commodities between which a consumer is indifferent.
- The slope of indifference curve is known as MRS.
- **Indifference Map:** It refers to a set of indifference curves corresponding to different income levels of the consumers. A collection of Indifference curve on a graphics called Indifference map. An indifference curve which is to the right and above another indifference curve corresponds to higher level of income and therefore, represents higher level of satisfaction.
- **Assumptions:**

1. The consumer acts rationally so as to maximise satisfaction.
2. There are two goods X and Y.
3. The prices of the two goods are given.
4. He prefers more of X to less of Y or more of Y to less of X.
5. The consumer arranges the two goods in a scale of preference which means that he has both 'preference' and 'indifference' for the goods.
6. Both preference and indifference are transitive. It means that if combination A is preferable to B, and B to C, then A is preferable to C. Similarly, if the consumer is indifferent between combinations A and B, and B and C, then he is indifferent between A and C. This is an important assumption for making consistent choices among a large number of combinations.
7. The consumer is in a position to order all possible combinations of the two goods.
8. Consumer's preferences becomes monotonic if the consumer between various bundles of two goods, prefers the bundle which has more of at least one of the goods and no less of other goods as compared to the other bundle.

- **Properties or Characteristics of Indifference Curves:**

1. It slopes downwards from left to right.
2. Indifference Curves are convex to the origin.
3. Indifference Curves will never intersect each other.
4. A higher Indifference Curve represents higher level of satisfaction.
5. Indifference Curve neither touches X-axis nor Y-axis.

- **Slope of an Indifference Curve:** An indifference curve slopes downward from left to right. In other words, an indifference curve is negatively sloped.
- **Marginal Rate of Substitution (MRS):** Marginal rate of substitution refers to the rate at which consumer is willing to give up amount of other goods to obtain one extra unit of the good in question without affecting total satisfaction. So, the rate of substitution of one commodity for another is called marginal rate of substitution. It is expressed as: $\frac{\Delta Y}{\Delta X}$ OR $\frac{\text{Sacrifice}}{\text{Gain}}$
- **Law of Diminishing Marginal Rate of Substitution (MRS):** This law states that as more and more units of Good-X are consumed, the consumer will sacrifice less and less units of Good-Y for each additional unit of Good-X. In other words, MRS falls continuously.
  For instance, a consumer is willing to give up 3 units of clothes for 1 unit of food when he moves from one combination to another. Then when he moves to yet another combination, he is willing to give up 2 units of clothes for 1 unit of food.
- **Consumer Budget:** It refers to the purchasing power of the consumer with which the consumer can buy various combinations of two goods.
- **Budget Set:** It refers to attainable combinations of set of two goods, given the prices of goods and income of the consumer.
- **Budget Line:** It is a line showing different possible combinations of two goods, which a consumer can buy given his budget and the prices of the goods. It is expressed as:

$$P_X Q_X + P_Y Q_Y = \text{Income of the consumer}$$

- **Price Ratio or Slope of Budget Line:** It is the rate at which the consumer is willing to exchange Goods-Y for Good-X. It is calculated as $\frac{P_X}{P_Y}$. The price ratio is represented by a negatively sloped straight line.
- **Consumer's Equilibrium:** It is the point at which the consumer derives maximum level of satisfaction within his budget.
- **Conditions of Consumer's Equilibrium:**

(i) Price Line should be tangent to Indifference Curve, that is, $MRS_{XY} = \frac{P_X}{P_Y}$

(ii) At the point of equilibrium, Indifference Curve is convex to the origin.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Define diminishing marginal rate of substitution? [OEB]

Ans. Marginal rate of substitution refers to the rate at which consumer is willing to give up amount of other goods to obtain one extra unit of the good in question without affecting total satisfaction. So, the rate of substitution of one commodity for another is called marginal rate of substitution.

Q.2. State whether the given statement is true or false with reason: [OEB]

(a) "Consumer's equilibrium based on the indifference curve analysis shows that marginal rate of substitution should be increasing."

Ans. False

Reason: Consumer's equilibrium based on the indifference curve analysis shows that marginal rate of substitution should be decreasing. Because diminishing marginal rate of substitution states that with the increase in consumption of one goods, consumption of other goods falls at decreasing rate.

(b) "One of the properties of indifference curve is that indifference curve should not intersect each other." [OEB]

Ans. True

Reason: The above given statement is true because if indifference curves intersect each other then there would be some combination on which both the indifference curves would lie but that is not the case because higher indifference curve gives higher satisfaction and lower IC shows lower satisfaction.

Q.3. Give the meaning of Indifference set. [OEB]

Q.4. If a person prefers bundle A(2,4) to B(2,3), then is he following the monotonic preferences? [OEB]

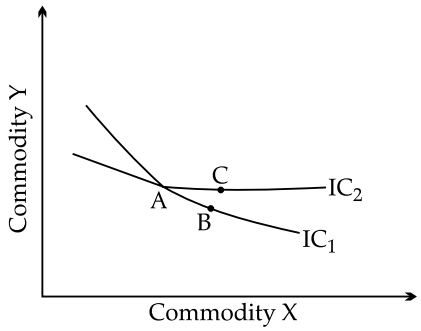
Ans. If a person prefers A(2,4) to B(2,3), then even though consumer has same unit of first commodity but less unit of other commodity in bundle B but more of it in bundle A. Hence, consumer will prefer bundle A to bundle B. thus, this shows that consumer follows monotonic preferences.

Q.5. If A is preferred to B and B is preferred to C, Ram says that A is preferred to C. Is he correct? [OEB]

Ans. Yes he is correct. Both preference and indifference are transitive. It means that if combination A is preferable to B, and B to C, then A is preferable to C. Similarly, if the consumer is indifferent between combinations A and B, and B and C, then he is indifferent between A and C. This is an important assumption for making consistent choices among a large number of combinations.

2 Marks Questions

Q.1.



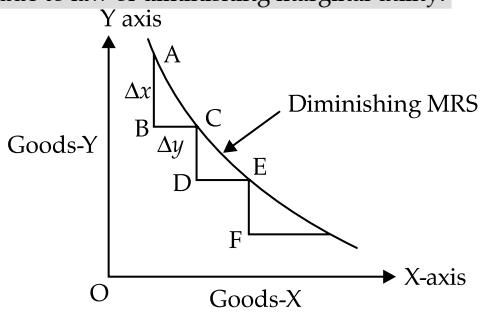
Why is the graph shown above not a common phenomenon in the consumer preferences?

[ISC Specimen Paper, 2023-24]

Ans. The graph shows that the two indifference curve IC_1 and IC_2 are intersecting each other, which cannot be the case as no two indifference curve can intersect each other as higher indifference curve will show higher level of satisfaction and in the above figure at point A both the indifference curves show the same level of satisfaction which is not possible. Thus, the above graph is not a common phenomenon in the consumer preferences.

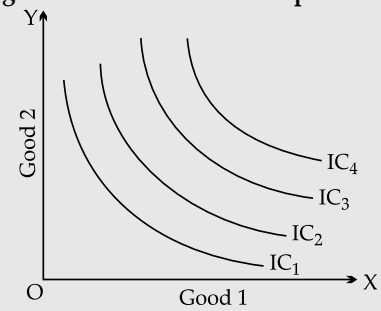
Q.2. Explain how DMRS is responsible for convexity of Indifference Curve. [ISC Specimen Paper, 2023-24]

Ans. An Indifference Curve will ordinarily be convex to the origin. This is because of Diminishing Marginal Rate of Substitution. It implies that as the consumer substitutes X for Y, the Marginal Rate of Substitution between them goes on diminishing. MRS diminishes because the utility of the goods consumed more falls due to law of diminishing marginal utility.



Q.3. Explain an indifference map, with the help of a diagram. [ISC 2019]

Ans. Diagram of an indifference map:



Indifference map is a group of indifference curves each one of which represents a given level of satisfaction. **[ISC Marking Scheme, 2019]**

Examiner's Comment

- Majority of the candidates did not draw the indifference map to explain it. Several candidates defined indifference curve instead of indifference map. In many answer scripts, the labelling of the indifference map was incorrect.

Answering Tip

- The students need to draw one curve for an indifference curve and at least three indifference curves while defining an indifference map.

Q.4. What is meant by budget line? [ISC 2016]

Q.5. Two indifference curves never intersect each other. Justify the statement. [ISC Specimen Paper, 2023]

Ans. As two indifference curves cannot represent the same level of satisfaction, they cannot intersect each other. It means, only one indifference curve will pass through a given point on a indifference map. They do not intersect because at the point of tangency, the higher curve will give as much as of the next commodities as is given by the lower indifference curve.

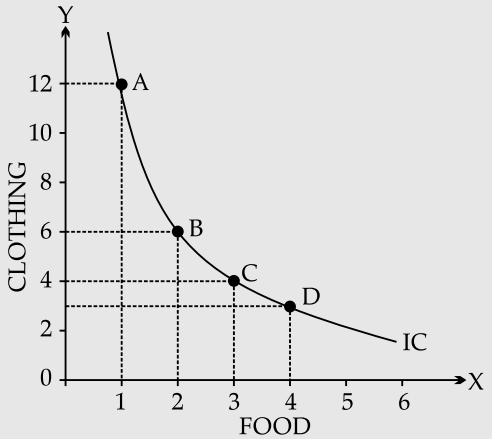
This question is for practice and its solution is available at the end of the chapter

3 Marks Questions

Q.1. Explain the concept of Diminishing Marginal Rate of substitution and show how it affects the indifference curve. [ISC 2020]

Ans. Diminishing Marginal Rate of Substitution:

- DMRS (X for Y) shows as consumer consumes more of X for Y, he will be willing to sacrifice lesser quantity of Y with every unit increase in consumption of X.
- IC curve is convex due to DMRS.



Units of Food	Units of Clothing	DMRS
1	12	--
2	6	6:1
3	4	2:1
4	3	1:1

As consumer moves from points A to D for every 1 unit increase in consumption of X lesser and lesser amount of Y is sacrificed.

(Concept of DMRS may also be given with the help of a schedule)

(Concept, Schedule and Relevance to be given)

[ISC Marking Scheme, 2020]

Examiner's Comment

- Most candidates wrote the definition of MRS instead of explaining the concept of Diminishing Marginal Rate of Substitution which not only made their answer incomplete but also incorrect as the they were unable to show how Diminishing MRS affects the indifference curve. Many candidates simply drew an indifference curve instead of showing the effect of DMRS on indifference curve. Candidates also did not draw a table to show the DMRS. A large number of candidates, instead of convexity, accorded the negative slope of the indifference curve to DMRS. In many answer scripts, diagrams were not properly scaled to show DMRS and hence the convex IC.

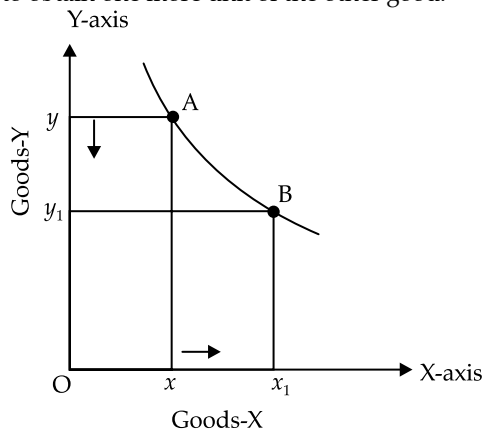
Answering Tip

- The difference between MRS and DMRS should be clear to students. Students should know that the effect of DMRS can be shown with the help of both, a schedule and a curve. They should be told that DMRS is responsible for the convexity of the indifference curve. For questions like this it is advisable that students draw a table and then write the explanation. They should first derive the DMRS with the help of a table and then show the ratio of substitution.

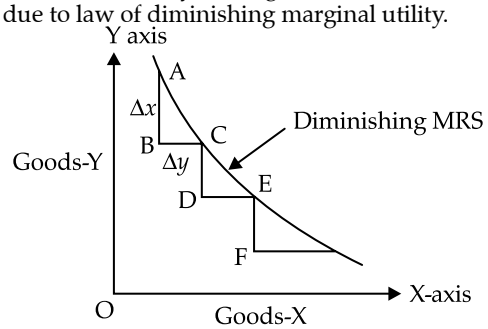
Q.2. Explain any two characteristics of an indifference curve. A I [ISC 2017]

Ans. The characteristics of an Indifference curve are:

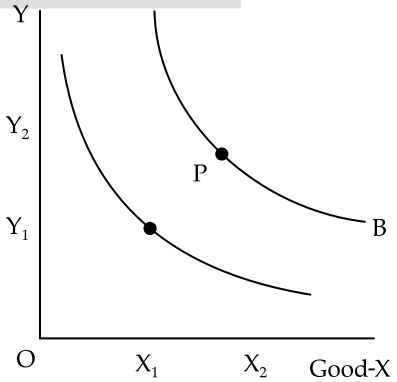
(i) Indifference Curve is downward sloping: Indifference Curve is downward sloping as when the consumer buys more units of one commodity he has to sacrifice the consumption of the other commodity so as to attain the same level of satisfaction. This means that the Marginal Rate of Substitution makes the indifference curve to be downward sloping. Marginal Rate of Substitution refers to the rate at which the consumer is willing to sacrifice one good to obtain one more unit of the other good.



(ii) Indifference Curve is Convex to the origin: An Indifference Curve will ordinarily be convex to the origin. This is because of Diminishing Marginal Rate of Substitution. It implies that as the consumer substitutes X for Y, the Marginal Rate of Substitution between them goes on diminishing. MRS diminishes because the utility of the goods consumed more falls due to law of diminishing marginal utility.

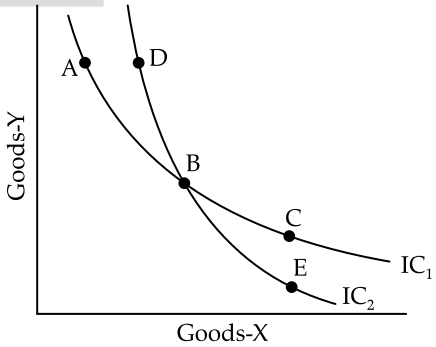


(iii) Higher the indifference Curve Higher is the level of satisfaction: Higher Indifference Curve represents higher level of satisfaction. In other words, any combination that lies on a higher Indifference Curve i.e., away from origin represents higher level of satisfaction. This is based on the assumption of monotonic preferences which means that greater consumption of a commodity by the consumer gives higher level of satisfaction.



(iv) Indifference curve never touch any axis: In the case of indifference curve, the curve never touch any axis due to the assumption that the consumer does always choose a combination of t is wo goods. If the indifference curve touches the x-axis, that means the consumer will not but the commodity X and if it touches the y-axis the consumer will not buy commodity Y.

(v) Indifference curve never intersect each other: Each Indifference Curve represents a different level of satisfaction.



[Any Two]

Examiner's Comment

- Many candidates did not explain the various characteristics of indifference curve. A number of candidates wrote the captions without any explanation.

Answering Tip

- Learn the characteristics of indifference curve with logical reasoning.

Q.3. Discuss two differences between cardinal utility and ordinal utility. [ISC 2016]

Ans.

S. No.	Basis	Cardinal utility	Ordinal utility
1.	Measurement	Utility is measured in terms of absolute numbers	Utility is measured in terms of scale of preference
2.	Equilibrium	$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} = MU^M$	$MRS_{xy} = \frac{P_x}{P_y}$
3.	Principle	Diminishing marginal utility	Diminishing marginal rate of substitution

[ISC Marking Scheme 2016]

Examiner's Comment

- Differences between cardinal and ordinal utility: The differences between cardinal utility and ordinal utility were not given clearly, e.g. Cardinal Utility – Can be measured; Ordinal Utility – Cannot be measured – this was incomplete.

Answering Tip

- Understand the meaning and concept of "cardinal"; and "ordinal".

Q.4. Discuss any two properties of indifference curve. [ISC 2015]

Q.5. What is a budget line? Why is it downward sloping? [OEB]

Ans. Budget Line is the locus of points that show such combinations of two goods that the consumer can buy with his given income and at given prices.

It is downward sloping because when consumer purchase more of one goods he must reduce the purchase of other goods with his given income and at given prices.

Q.6. If $MRS_{xy} = 3$, $P_x = 3$ and $P_y = 3$, then is the consumer in equilibrium according to the ordinal utility analysis? [OEB]

This question is for practice and its solution is available at the end of the chapter

Ans. Here,

$MRS_{XY} = 3$

$P_X/P_Y = 3/3 = 1$

Hence, MRSxy > P_X/P_Y, i.e.

3>1

Hence, the rate at which consumer wants to substitute goods X for goods Y is greater than the rate at which market is willing to exchange. Hence consumer consumes more of X and less of Y until MRSxy becomes equal to P_X/P_Y.

6 Marks Questions

Q.1. Explain how a consumer attains equilibrium using indifference curve analysis. [ISC 2019]

OR

Explain the concept of consumer's equilibrium with the help of indifference curve analysis. [ISC 2016]

Ans. A consumer attains equilibrium when he **maximises** his total utility with his given income and prices of the two commodities. In terms of indifference curve analysis, he attains equilibrium when he can reach the highest possible indifference curve with his given budget constraint. Equilibrium conditions

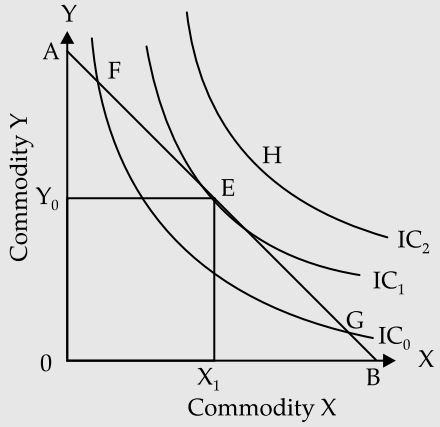
Necessary condition:

Slope of IC = Slope of the budget line:

$$MRSxy = \frac{P_x}{P_y}$$

Sufficient condition:

The IC is **convex** to the origin at the point of equilibrium.



The consumer attains equilibrium at point E where,

Slope of IC = Slope of the budget line

The **point of tangency** of the highest possible IC i.e., IC_1 and the budget line AB gives the consumer the highest possible satisfaction.

If the consumer purchases any combination like F or G on IC_0, he will get lesser satisfaction compared to point E.

The consumer cannot purchase any combination like H on IC_2, because it is out of reach of the consumer.

∴ The consumer attains equilibrium at point E.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Many candidates did not write the meaning of equilibrium. The budget line in the graph drawn by several candidates did not touch the axes. Also, in many answer scripts, the tangent did not touch the indifference curve. Some candidates drew only one or two ICs whereas all three were to be drawn. Some candidates seemed to be confused with the law of equi-marginal utility approach. The conditions of equilibrium were not written by a few candidates.

Answering Tip

- The students need to understand that they should write the meaning of consumer's equilibrium, the conditions of equilibrium, the graph must include the indifference map and various points should be considered to justify the point of equilibrium.
- The students need to justify their answers.

Q.2 Explain why an indifference curve is (a) downward sloping and (b) convex. [OEB]

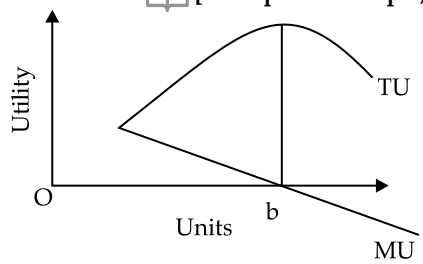
Ans. (i) Indifference Curve is downward sloping: Indifference Curve is downward sloping as when the consumer buys more units of one commodity he has to sacrifice the consumption of the other commodity so as to attain the same level of satisfaction. This means that the Marginal Rate of Substitution makes the indifference curve to be downward sloping. Marginal Rate of Substitution refers to the rate at which the consumer is willing to sacrifice one good to obtain one more unit of the other good.

(ii) Indifference Curve is Convex to the origin: An Indifference Curve will ordinarily be convex to the origin. This is because of Diminishing Marginal Rate of Substitution. It implies that as the consumer substitutes X for Y, the Marginal Rate of Substitution between them goes on diminishing. MRS diminishes because the utility of the goods consumed more falls due to law of diminishing marginal utility.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. Refer to the graph given below and determine the behaviour of TU as MU decreases up to the point 'b'. **[ISC Specimen Paper, 2023-24]**



(A) TU increases up to a limit and remains positive.
(B) TU increases at an increasing rate
(C) TU increases at a diminishing rate.
(D) TU is at a point of satiety.

Ans. Option (C) is correct.

Explanation: Up to the point 'b', the TU increases at a diminishing rate and reaches its maximum point. Beyond the point TU starts to fall.

Q.2. The aggregate utility obtained from the consumption of a specific unit of a commodity is called: **[ISC 2023]**

(A) Total utility **(B)** Total utility
(C) Maximum utility **(D)** Additional utility

Ans. Option (B) is correct.

Explanation: Total utility refers to the aggregate or total amount of satisfaction that a rational consumer receives through the consumption of a specific good or service.

Q.3. Convexity of indifference curve implies _____ marginal rate of substitution. **[ISC Semester-I, 2022]**

(A) Increasing **(B)** Decreasing
(C) Constant **(D)** Zero

Ans. Option (B) is correct.

Explanation: An Indifference Curve will ordinarily be convex to the origin. This is because of Diminishing Marginal Rate of Substitution. It implies that as the consumer substitutes X for Y, the Marginal Rate of Substitution between them goes on diminishing.
MRS diminishes because the utility of the goods consumed more falls due to law of diminishing marginal utility.

Q.4. The indifference curve means: **[ISC Semester-I, 2022]**

(A) Equal consumption of two goods.
(B) Equal utility from the consumption of combination of two goods.
(C) Equal income of the consumer.
(D) Equal prices of goods consumed.

Q.5. Additional utility derived from the consumption of an additional unit of a commodity is called: **[ISC Semester-I, 2022]**

(A) Average utility **(B)** Marginal Utility
(C) Total Utility **(D)** Additional Utility

Ans. Option (B) is correct.

Q.6. What is the formula for calculating M.U.? **[ISC Specimen Paper, Semester-I, 2021-22]**

(A) $TU_N + TU_{N+1}$ **(B)** $TU_N - TU_{N+1}$
(C) $TU_N - TU_{N+1}$ **(D)** $TU_N - TU_{N-1}$

Ans. Option (D) is correct.

Q.7. The slope of the indifference curve is equal to: **[ISC Specimen Paper, Semester-I, 2021-22]**

(A) One.
(B) Marginal utility.
(C) Marginal rate of substitution
(D) None of these.

Ans. Option (C) is correct.

Q.8. Which situation is consistent with the law of diminishing marginal utility? **[OEB]**

(A) The more pizza Henry eats, the more he enjoys another slice.
(B) The more pizza Henry eats, the less he enjoys another slice.
(C) Henry's marginal utility from eating pizza becomes positive after eating three slices.
(D) Henry's marginal utility from eating pizza reaches a maximum when total utility is zero.

Ans. Option (B) is correct.

Explanation: As per the law of diminishing marginal utility, the marginal utility derived from the consumption of one additional unit of a commodity decreases. This is best explained in option (b).

Q.9. A budget constraint line is a result of: **[OEB]**

(A) Market price of X.
(B) Market price of Y.
(C) Income of the consumer.
(D) Revenue of the producer.

Q.10. Which of the following shows the various combinations of two products that give same amount of satisfaction? **[OEB]**

(A) Budget line. **(B)** Budget set.
(C) Indifference set. **(D)** Indifference curve.

Ans. Option (D) is correct.

Explanation: Indifference curve is the locus of point that represent different combinations of two goods that give the same satisfaction to the consumer.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

This question is for practice and its solution is available at the end of the chapter

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): Law of diminishing marginal utility states that as more and more units of a commodity are consumed, marginal utility derived from every additional unit must decline.
Reason (R): When MU is negative, TU will be decreasing. **[OEB]**

Ans. Option (B) is correct.
Explanation: The Assertion is the statement of the law and doesn't need any reason for it.

Q.2. Assertion (A): Marginal Utility diminishes as the person consumes more unit of a commodity.
Reason (R): Utility derived from each unit of a commodity is less than that of the previous one. **[OEB]**

Ans. Option (A) is correct.
Explanation: Marginal Utility is derived from the consumption of one extra unit of a commodity. Thus, as we get lesser utility from every next unit of commodity consumed, thus the marginal utility of the commodity decreases.

Q.3. Assertion (A): Rare commodities like antiques do not follow the law of diminishing marginal utility.
Reason (R): Law of marginal utility states that as more and more units of a commodity are consumed, marginal utility derived from additional unit must decline. **[OEB]**

Ans. Option (B) is Correct.
Explanation: As rare commodities utility increases as we purchase more of it, thus it does not follow the law of diminishing marginal utility.

Q.4. Assertion (A): Indifference Curves are always convex to the origin.
Reason (R): Diminishing Marginal Rate of Substitution. **[OEB]**

Ans. Option (A) is correct.
Explanation: The indifference curve is convex to the origin because of the diminishing Marginal Rate of Substitution.

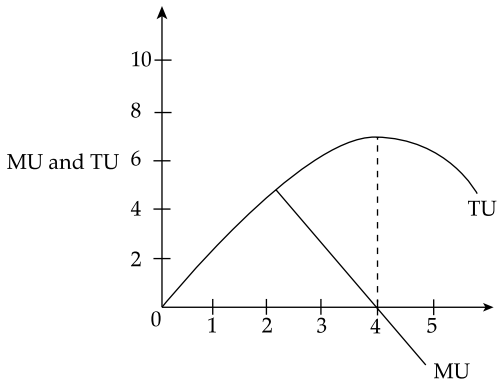
Q.5. Assertion (A): Conditions of consumer equilibrium is Marginal Rate of Substitution (MRS) = $\frac{P_X}{P_Y}$ and MRS falls as more is consumed of X.
Reason (R): The consumer is said to be in equilibrium when he spends income in such a way that he gets maximum satisfaction. **[OEB]**

Ans. Option (A) is correct.
Explanation: At the consumer equilibrium, the consumer spends income in such a way that his utility is maximised as the Marginal rate of substitution falls, if more unit of X is consumed.

Case Based Questions

[A] Multiple Choice Questions

I. Study the diagram given below and answer the following questions that follows: **[ISC Semester-I, 2022]**



(a) When TU is maximum, then MU is:
(A) Zero **(B)** Maximum
(C) Positive **(D)** Negative

Ans. Option (A) is correct.

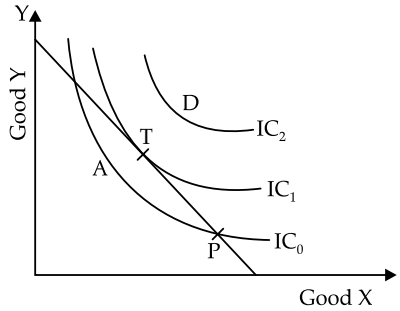
(b) When TU decreases MU is _____.
(A) Increasing **(B)** Decreasing
(C) Negative **(D)** Zero

Ans. Option (C) is correct.

(c) When TU is increasing at a diminishing rate then MU is _____.
(A) Decreasing but is positive.
(B) Increases
(C) Negative **(D)** Becomes S Shaped.

Ans. Option (A) is correct.

II. Study the diagram given below and answer the questions that follows: **[ISC Semester-I, 2022]**



(a) In the diagram above, the point of consumer equilibrium is:
(A) D **(B)** T **(C)** A **(D)** B

Ans. Option (B) is correct.

(b) The point which is unattainable is:
(A) A **(B)** T **(C)** B **(D)** D

Ans. Option (D) is correct.

(c) The condition for consumer's equilibrium is:
(A) $MU_X = P_X$ **(B)** $\frac{MU_X}{P} = \frac{MU_Y}{P}$
(C) MC = MR **(D)** $MRS_{XY} = \frac{P_X}{P_Y}$

Ans. Option (D) is correct.

[B] Subjective Based Questions

I.

Pens (units)	1	2	3	4	5
MU (units)	25	20	15	10	5

(a) What law does the above table show? [OEB]

Ans. The table shows the law of diminishing marginal utility.

(b) State the law.

Ans. Law of diminishing marginal utility states that as more and more units of a commodity are consumed, marginal utility derived from every additional unit must decline.

(c) What are the assumptions on which the law is based?

Ans. This law is based on these assumptions:

(i) The utility can be measured in terms of units
(ii) The marginal utility of money is constant
(iii) The consumption must be in identical units
(iv) The consumption is continuous
(v) The consumption is made in terms of standard units
(vi) The consumer is rational
(vii) The income of the consumer and the prices of goods are constant. **[Any Four] [4 × ½ = 2]**

(d) What will be the total utility of the pen at various level?

Ans.

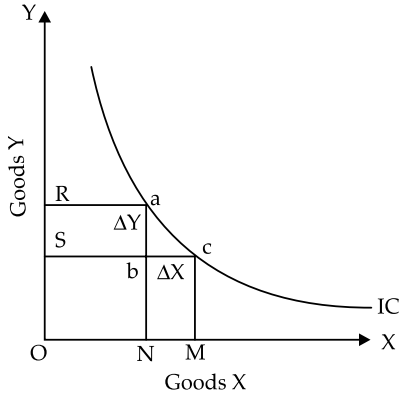
Pens (Units)	MU	TU
1	25	25
2	20	45
3	15	60
4	10	70
5	5	75

(e) State the relationship between Total Utility and Marginal Utility.

Ans. The relationship between Total Utility and Marginal Utility are:

(i) When MU is positive, TU is increasing.
(ii) When MU is zero, TU is maximum.
(iii) When MU is negative, TU is decreasing

IV. [OEB]



(a) What does the figure show? Explain the thing shown in the curve.

Ans. The figure shows the Marginal Rate of Substitution in the Indifference Curve. Marginal Rate of Substitution refers to the rate at which the consumer is willing to sacrifice one goods to obtain one more unit of the other goods.

(b) Define the curve shown.

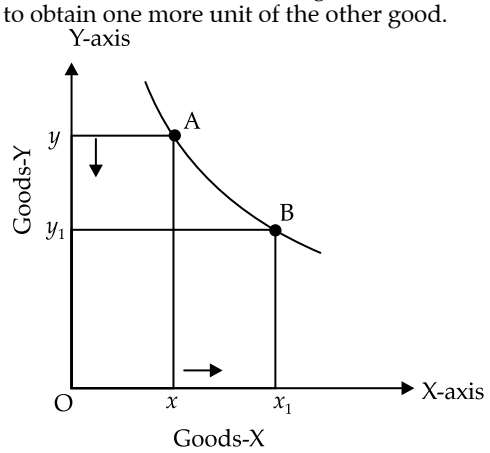
Ans. Indifference Curve is the locus of point that represent different combinations of two goods that give the same satisfaction to the consumer.

(c) Explain any two properties of the curve shown in the figure.

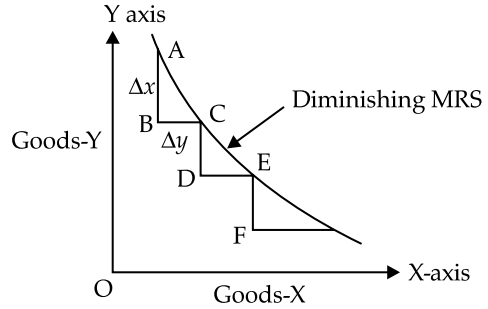
Ans. Refer to question Q2 of 3 Mark Question Topic 2.

Ans. The characteristics of an Indifference curve are:

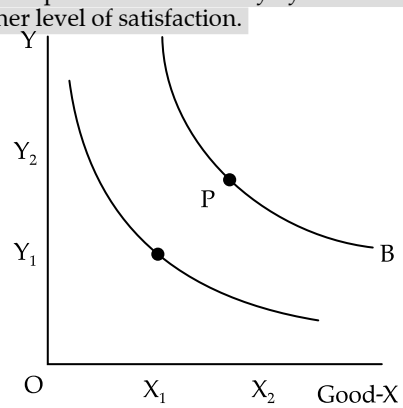
(i) Indifference Curve is downward sloping: Indifference Curve is downward sloping as when the consumer buys more units of one commodity he has to sacrifice the consumption of the other commodity so as to attain the same level of satisfaction. This means that the Marginal Rate of Substitution makes the indifference curve to be downward sloping. Marginal Rate of Substitution refers to the rate at which the consumer is willing to sacrifice one good to obtain one more unit of the other good.



(ii) Indifference Curve is Convex to the origin: An Indifference Curve will ordinarily be convex to the origin. This is because of Diminishing Marginal Rate of Substitution. It implies that as the consumer substitutes X for Y, the Marginal Rate of Substitution between them goes on diminishing. MRS diminishes because the utility of the goods consumed more falls due to law of diminishing marginal utility.

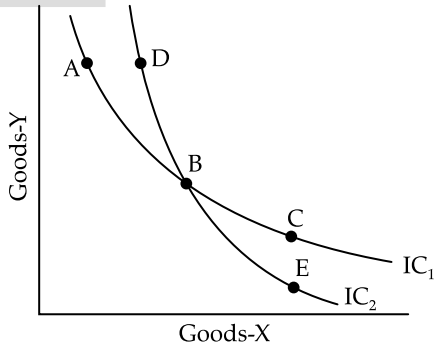


(iii) Higher the indifference Curve Higher is the level of satisfaction: Higher Indifference Curve represents higher level of satisfaction. In other words, any combination that lies on a higher Indifference Curve i.e. away from origin represents higher level of satisfaction. This is based on the assumption of monotonic preferences which means that greater consumption of a commodity by the consumer gives higher level of satisfaction.



(iv) Indifference curve never touch any axis: In the case of indifference curve, the curve never touch any axis due to the assumption that the consumer does always choose a combination of t is wo goods. If the indifference curve touches the x-axis, that means the consumer will not but the commodity X and if it touches the y-axis the consumer will not buy commodity Y.

(v) Indifference curve never intersect each other: Each Indifference Curve represents a different level of satisfaction.



[Any Two]

Examiner's Comment

- Many candidates did not explain the various characteristics of indifference curve. A number of candidates wrote the captions without any explanation.

(d) What are the assumptions on which the above curve is based on?

Ans. The assumption on which the above curve is based on are:

(i) The consumer acts rationally so as to maximise satisfaction.

(ii) There are two goods X and Y.

(iii) The prices of the two goods are given.

(iv) He prefers more of X to less of Y or more of Y to less of X.

(v) The consumer arranges the two goods in a scale of preference which means that he has both 'preference' and 'indifference' for the goods.

(vi) Both preference and indifference are transitive. It means that if combination A is preferable to B, and B to C, then A is preferable to C. Similarly, if the consumer is indifferent between combinations A and B, and B and C, then he is indifferent between A and C. This is an important assumption for making consistent choices among a large number of combinations.

(vii) The consumer is in a position to order all possible combinations of the two goods.

(viii) Consumer's preferences becomes monotonic if the consumer between various bundles of two goods, prefers the bundle which has more of at least one of the goods and no less of other goods as compared to the other bundle. **[Any Two]**

(e) If more than one such curve is drawn, what is it called?

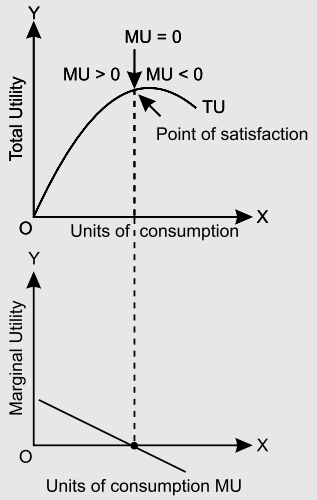
Ans. If more than one curve is drawn it is called an indifference map.

It refers to a set of indifference curves corresponding to different income levels of the consumers. An indifference curve which is to the right and above another indifference curve corresponds to higher level of income and therefore, represents higher level of satisfaction.

Solutions for Practice Questions (Topic-1)

2 Marks Questions

3. Marginal Utility: Utility derived from every additional unit by the consumer is marginal utility. At one point, consumer's total utility becomes maximum and then his total utility falls. This is the point where marginal utility becomes negative.



[ISC Marking Scheme 2015]

Examiner's Comment

- Instead of change in utility, candidates wrote change in commodity which as a concept was incorrect. Some candidates wrote that marginal utility becomes negative when total utility is maximum, which is correct.

Answering Tip

- Explain relationship between TU and MU with the help of graphs and schedule.

Solutions for Practice Questions (Topic-2)

1 Mark Questions

3. Indifference set can be referred to as a graph of various indifference curves. Each indifference curve represents different level of satisfaction.

2 Marks Questions

4. It is a negatively sloped line representing the price ratio of two goods that can be purchased in a consumer's given income. It is a straight line as the prices of the two commodities are assumed to be given while drawing it.

[ISC Marking Scheme 2016]

Examiner's Comment

- Both the parameters – 'prices of two goods' and 'income of the consumer's need to be mentioned while defining budget line. Many candidates wrote only one parameter.

Answering Tip

- Learn the importance of 'prices of two goods' and 'income of the consumer' and how these two parameters can alter the position of the budget line.

3 Marks Questions

4. **Properties of Indifference Curve:**

(i) Downward sloping due to marginal rate of substitution or in order to increase additional units of one, other commodity has to be substituted.

(ii) It is convex to origin – showing decreasing MRS – consumer is ready to substitute or sacrifice less as he increases consumption of one commodity.

(iii) Higher indifference curves gives higher level of satisfaction – as he consumes more of both the commodities.

(iv) Two indifference curves never intersect.

(Any two) [ISC Marking Scheme 2015]

Solutions for Objective Type Questions

[A] Multiple Choice Questions

4. **Option (B) is correct.**

Explanation: Indifference Curve is the locus of point that represent different combinations of two goods that give the same satisfaction to the consumer.

9. **Option (C) is correct.**

Explanation: Budget Line or Budget constraint line is the locus of points that show such combinations of two goods that the consumer can buy with his given income and at given prices. It is downward sloping because when consumer purchase more of one goods, he must reduce the purchase of other goods with his given income and at given prices.

REFLECTION

(a) Can you explain the relationship between Total utility and Marginal Utility?

(b) Do you understand the concept of indifference curve and its characteristics?

(c) How does a consumer attain their equilibrium?

(d) Do you understand the law of diminishing marginal utility and Equi-marginal utility?